

RAG Medical Reranker - Retrieval-Hard Benchmark Report Draft

Hard-level experiment narrative and current results. BGE section is intentionally left as a placeholder.

CPS3830 Machine Learning

RAG Reranker

Hard Benchmark

1. Executive Summary

This report focuses on the retrieval-hard benchmark of the **RAG_Medical_Reranker** project. The original controlled benchmark successfully verified the full reranking pipeline, but it was too easy: the embedding baseline was already close to perfect. To make the evaluation more realistic, we constructed a harder benchmark where most negative candidates are retrieved by embedding similarity rather than randomly sampled.

Current key finding: On the hard benchmark, the embedding baseline drops to **0.886 P@1**, while the supervised feature-based ML reranker reaches **0.998 P@1**. This confirms that the hard benchmark exposes the value of supervised reranking in a realistic RAG top-k setting.

The BCE cross-encoder has also been evaluated. BCE does not improve P@1 over the embedding baseline, but it improves MRR and NDCG@5, indicating better top-k robustness. The BGE result is still pending and should be inserted later.

2. Why We Added a Retrieval-Hard Benchmark

The controlled candidate set was useful as a pipeline sanity check, but it did not fully stress the reranking problem. In that setting, many negatives were random or only weakly related, so cosine embedding similarity could already rank the positive answer at the top in nearly all cases.

Benchmark	Candidate design	Main purpose	Limitation / value
Controlled benchmark	1 positive + same-department/ random negatives	Verify end-to-end pipeline and unified output format	Too easy; embedding baseline nearly saturated
Retrieval-hard benchmark	1 positive + 7 embedding-hard negatives + 2 random negatives	Simulate realistic RAG top-k retrieval	More difficult; better reveals reranker value

The new benchmark is not meant to replace the controlled benchmark. Instead, the two benchmarks serve different roles: the controlled benchmark proves that the system works; the hard benchmark evaluates whether reranking is actually useful under difficult retrieval conditions.

3. Hard Benchmark Construction

The hard benchmark was generated as `data/processed/candidates_hard.csv`. It contains 500 medical questions and 5,000 query-candidate pairs.

Property	Value
Number of questions	500
Total query-candidate pairs	5,000
Candidates per question	10
Positive per question	1
Embedding-hard negatives per question	7
Random negatives per question	2

The generation process first samples medical questions from the raw pediatric and oncology QA data. For each question, the original answer is used as the only positive candidate. The hard negatives are selected by embedding similarity: they are contexts that the embedding retriever considers highly similar to the question, but are not the original gold answer. This mimics a realistic first-stage RAG retriever, where the top-k candidates are often semantically close and difficult to distinguish.

Important design choice: We did not use 9 embedding-hard negatives. We kept 2 random negatives per question to avoid making the benchmark overly ambiguous and to preserve a small stability anchor in each candidate set.

4. Methods Evaluated So Far

Method	Input representation	Training / supervision	Role in the experiment
embedding_baseline	Precomputed embedding similarity score	Unsupervised	First-stage retriever baseline
embedding_department_bonus	Embedding score + same-department bonus	Unsupervised heuristic	Domain-rule ablation
lightweight_ml	Engineered retrieval features	Supervised logistic regression	Feature-based learned reranker
BCE	Raw question-context text pair	Zero-shot pretrained cross-encoder	Transformer reranker comparison
BGE	Raw question-context text pair	Zero-shot pretrained reranker	Pending

The lightweight ML model should be interpreted carefully. It is not a raw-text semantic model like BCE. It is a supervised feature-based reranker that learns how to combine retrieval-oriented signals such as embedding score, keyword overlap, department match, and length features. Therefore, the comparison should not be framed as "logistic regression understands medical language better than BCE." A more accurate interpretation is that, under this fixed RAG retrieval distribution, supervised feature calibration is highly effective.

5. Main Results on the Hard Benchmark

Method	Num QIDs	P@1	P@3	P@5	MRR	NDCG@5	Avg latency ms/pair
embedding_baseline	500	0.886	0.968	0.986	0.9285	0.9416	0.0000
embedding_department_bonus	500	0.886	0.970	0.986	0.9290	0.9420	0.0000
lightweight_ml	500	0.998	1.000	1.000	0.9990	0.9993	0.0004
BCE	500	0.884	0.988	0.998	0.9358	0.9516	0.5252
BGE	Pending	TBD	TBD	TBD	TBD	TBD	TBD

6. Interpretation of the Current Results

6.1 The hard benchmark succeeded

The embedding baseline dropped from near-perfect performance on the controlled benchmark to 0.886 P@1 on the retrieval-hard benchmark. This confirms that the new benchmark is meaningfully harder. The hard negatives are no longer trivial mismatches; they are semantically similar contexts retrieved by the embedding model itself.

6.2 Department bonus is not enough

The department bonus barely changes the result: P@1 remains 0.886 and MRR increases only from 0.9285 to 0.9290. This shows that the main challenge is not simply department mismatch. The difficult cases require finer-grained relevance modeling than a simple medical-domain heuristic.

6.3 Supervised feature-based ML is highly effective

The lightweight ML reranker improves P@1 from 0.886 to 0.998. This is a large improvement. The result suggests that a small supervised reranker can learn how to recalibrate retrieval signals under a fixed RAG candidate distribution. It is also extremely fast, with average latency of only 0.0004 ms per pair.

Careful wording: This does not prove that logistic regression has stronger semantic understanding than BCE. The ML model is supervised on the benchmark distribution and uses engineered retrieval features. BCE is a zero-shot pretrained text cross-encoder. The fair

interpretation is: supervised feature-based reranking can be highly effective and efficient for this specific RAG setting.

6.4 BCE improves top-k robustness but not top-1 accuracy

BCE has $P@1 = 0.884$, slightly below the embedding baseline. However, it improves $P@3$ from 0.968 to 0.988, $P@5$ from 0.986 to 0.998, MRR from 0.9285 to 0.9358, and NDCG@5 from 0.9416 to 0.9516. This means BCE often keeps the positive candidate near the top, but does not always place it at rank 1. This is an important nuance: BCE is not useless, but its benefit appears more in top-k robustness than in top-1 retrieval under this benchmark.

7. Pending BGE Section

BGE results should be inserted here after the group member finishes the hard benchmark run.

The final report should compare BCE and BGE as two off-the-shelf pretrained rerankers. Possible interpretations:

- If BGE outperforms BCE: discuss model-specific suitability for Chinese medical QA reranking.
- If BGE is similar to BCE: argue that pretrained cross-encoders improve top-k robustness but are not necessarily adapted to the current benchmark distribution.
- If BGE underperforms: frame it as evidence that zero-shot pretrained reranking may require domain adaptation or supervised fine-tuning for medical QA.

8. Next Analysis Needed: Error Analysis

To make the hard benchmark results more scientifically convincing, we should analyze individual ranking errors instead of relying only on aggregate metrics. The most important cases are:

- Cases where embedding_baseline fails at rank 1 but lightweight_ml succeeds.
- Cases where BCE fails at rank 1 but lightweight_ml succeeds.
- Cases where all available methods fail.
- Cases where embedding_baseline succeeds but BCE fails.

This analysis will help explain whether the ML reranker is correcting meaningful retrieval errors or merely fitting benchmark artifacts. It can also reveal whether some hard negatives are partially relevant or ambiguous.

9. Next Analysis Needed: Feature Importance

Because the lightweight ML result is very strong, we need to inspect which features drive its predictions. The target output should be:

```
results_hard/ml_feature_importance.csv
```

It should contain logistic regression coefficients for features such as embedding_score, keyword_overlap, department_match, question_length, context_length, and length_ratio. This will

clarify whether the ML reranker is mostly relying on embedding_score or whether additional features contribute to the performance gain.

10. Limitations

- The hard benchmark includes only two broad medical domains: pediatrics and oncology.
- Each question has only one labeled positive answer, although some medical QA negatives may be partially relevant.
- The lightweight ML model is trained on the benchmark distribution, while BCE is zero-shot; this difference must be acknowledged.
- The current evaluation measures evidence ranking quality, not final generated medical answer quality.
- BGE results are still pending, so the transformer reranker comparison is incomplete.

11. Report-Ready Conclusion

The retrieval-hard benchmark provides a more realistic and discriminative evaluation of RAG reranking than the original controlled benchmark. By replacing most random negatives with embedding-retrieved hard negatives, the task becomes substantially more difficult for the embedding baseline. The department bonus ablation shows that simple domain heuristics are not sufficient. The supervised feature-based ML reranker achieves the strongest current performance, suggesting that learned calibration of retrieval signals can be highly effective in a fixed RAG pipeline. BCE improves top-k robustness but does not improve top-1 ranking, highlighting the trade-off between pretrained semantic reranking and task-specific supervised reranking. The next step is to add BGE results and complete error analysis plus feature importance analysis to better explain why the supervised ML model performs so strongly.

Generated as a working draft for the hard-level benchmark. Update after BGE, error analysis, and feature importance results are available.